UNITED STATES FOREST SERVICE (USFS)
INCIDENT PROCUREMENT OPERATIONS (IPO)
REQUEST FOR INFORMATION (RFI)
UAS CAMERAS/SENSORS

SUBMITTAL OF INFORMATION:

This RFI is for market research only and does not constitute a solicitation for proposals. The information provided may be utilized by the USFS in developing its acquisition strategy and requirements documents. It does not commit the Government to pay any cost incurred in the submission of responses or in making necessary studies or designs, nor to contract for supplies or services. No funds have been authorized, appropriated, or received for this contemplated effort. Any cost incurred shall be at the Offeror's own risk. Live demonstrations that are currently integrated with approved USFS unmanned aircraft may be requested at a future date. Responses shall include potential Offeror's Unique Entity Identifier (UEI), be in PDF format, and emailed to the Contracting Officer by 1200 hours MT, June 12th, 2026, at steven.b.peterson@usda.gov.

DESCRIPTION:

The purpose of this notice is to obtain information regarding potential Offerors capable of providing various camera/sensor systems for use with approved USFS agency fleet for the purpose of mapping, surveying, providing situational awareness, high quality stills and video of our National Forests and Grasslands. The Government is seeking information regarding payloads that can be integrated with any of the following Unmanned Aircraft (UA):

- Freefly AltaX
- Freefly Astro

BACKGROUND:

The Government is interested in the use of system payloads to map and survey targeted areas of interest with the use of Unmanned Aircraft System (UAS). This may include resource management activities (vegetation mapping and characterization, flood plain mapping, erosion monitoring, hazard/disaster assessment, terrain analysis, heritage site mapping, etc.) as well as engineering and survey activities (infrastructure inspections, topographic surveys for infrastructure development, etc.). Provide written documentation of the capability to perform UAS data collection utilizing the different types of cameras/sensors. Offeror can respond to one payload or multiple payloads. Include UAS and payload specifications in the response. If equipment does not appear to meet minimum agency mission needs or integrations to approved USFS agency fleet, you will not be considered to perform the demonstration, should they occur. If you cannot perform the tasks, provide a description and rationale for what it may take to integrate the payload with the agency UA to accomplish the task (funding, additional time, additional resources such as multiple payloads, UAS, or pilots, etc.). The Government is interested in UAS and payload configurations that meet or exceed the requirements of this RFI. Any demonstration request will require current FAA-qualified remote pilot(s) controlling the aircraft. Proof of qualification will be required before approval of any demonstration.

TECHNICAL REQUIREMENT FOR FULL-FRAME PHOTOGRAPHY CAMERA:

This defines the desired technical, operational, environmental, and interface requirements for a next-generation, full-frame mapping camera compatible with Freefly aircraft utilizing Freefly's smart dovetail integration. The payload is intended for high accuracy photogrammetry, land management, forestry, wildfire recovery mapping, and engineering-grade survey missions.

Scope: Applies to the design, development, integration, and testing of a stabilized, full-frame photogrammetry camera system incorporating:

- Full-frame still-image sensor
- Interchangeable fixed-focus lenses
- Extremely fast mechanical shutter with actuation time <4 ms
- High-frequency, sustained capture capability (≤0.7 s interval)
- Integrated gimbal stabilization
- Freefly-compatible mechanical and electrical interfaces
- Global Navigation Satellite System (GNSS) time synchronization
- Motion Imagery Standards Board (MISB) compliant
- National Defense Authorization Act (NDAA) compliant
- Adjustable internal aperture range f/2.8 – f/22

System shall provide:

- Survey-grade photogrammetry
- High-resolution still capture
- Fast capture intervals for high-speed flight
- Wide-area coverage via full-frame sensor + 24 mm to 50mm lens options
- Deterministic shutter timing
- Freefly API/SDK integration

Functional Requirements:

- Complete installed payload(sensor + gimbal + mounting hardware + required accessories) weight shall be ≤ 1,100 g
- Power consumption shall be ≤30 W during capture
- Environmental protection shall be IP54 or higher
- Operating temperature shall be −20° C to 50° C
- Storage temperature shall be −20° C to 50° C
- Payload shall mount using the Freefly Quick-Detach Dovetail Payload Interface

Accuracy Requirements:

- Time Synchronization
  - Camera shall support microsecond-level synchronization with aircraft Global Navigation Satellite System/Inertial Measurement Unit (GNSS/IMU)
  - Timestamping shall be deterministic and externally verifiable

Camera Requirements:

- Sensor
    - Sensor shall be full-frame (36×24 mm)
    - Effective resolution shall be ≥ 60 megapixels
    - Pixel size shall be ≈ 4 µm
    - Sensor shall support **lo**w-noise, high-sensitivity imaging for dawn/dusk operations
- Shutter
    - Shutter shall be mechanical
    - Shutter speed shall support 1/4000 s or faster
    - **S**hutter timing shall be synchronized at the microsecond level with aircraft GNSS
- Capture Rate
    - Camera shall support ≤ 0.7 s interval between images during flight
    - Capture interval shall be maintained for duration of single flight
    - Camera shall maintain full resolution at maximum capture rate

Lens Requirements:

- The system shall support interchangeable fixed-focus lenses ranging from 24mm to 50mm
- Lenses shall maintain fixed focus at infinity for mapping
- Mount
    - Lens mount shall be camera compatible or industry standard (e.g., DL mount equivalent)
    - Mount shall support balancing rings or filters as needed

Gimbal Requirements:

- Stabilization
    - Gimbal shall provide 3 axis stabilization
    - Stabilization accuracy shall be ≤ ±0.03°
- Mechanical Range
    - Tilt: −90° to +30°
    - Roll: ±30°
    - Pan: ±320°
- Capture Modes
    - Gimbal shall support Smart Oblique Capture (automated multi-angle mapping)
    - Gimbal shall support Nadir locked mode for standard mapping

Storage and Data Requirements:

- Media
    - Camera shall support UHS-I V60 or faster SD cards

    - Maximum supported capacity shall be ≥ 512 GB
- File Types
    - Photo format shall be JPEG + RAW
    - The camera shall record JPEG imagery at a minimum quality setting of 90% (JPEG quality factor ≥90), with a compression ratio not exceeding 10:1, and without additional in-camera smoothing or sharpening beyond the manufacturer's minimal default
    - RAW image format shall support ≥12-bit files
    - Camera shall store GNSS raw observation data and oriented imagery metatdata for post processing
- File System
    - File System shall be exFAT
- Full Motion Video
    - The camera shall support recording Full Motion Video (FMV) streams with MISB compliant metadata, including but not limited to:
        - Platform position
        - Platform attitude
        - Sensor position
        - Sensor attitude
        - Sensor FOV (horizontal/vertical)
        - Sensor relative azimuth/elevation
        - Platform velocity (if available)
        - Frame center coordinates
        - Time stamps
        - Field of view
        - Slant range (if available)
        - Ground sample distance (if applicable)
    - Metadata shall conform to Motion Imagery Standards Board (MISB) standards appropriate for airborne ISR systems (e.g., ST 0601, ST 0903, or successor standards)
    - MISB metadata shall be embedded in the video stream in a manner compatible with common FMV exploitation tools used by public safety, federal land management, and SAR operations
    - The system shall support synchronized metadata for both real time streaming and onboard recording

Software and Integration Requirements:

- Freefly Integration
    - Camera shall be fully controllable via Freefly API/SDK

    - Camera shall support automated mapping missions through Freefly Ground Control
- Metadata
    - EXIF shall include:
        - GNSS position
        - GNSS time
        - Camera Orientation(heading(in degrees), pitch(in degrees), roll)
        - Camera field of view(HFoV/VFoV)
        - Focal length
        - Platform attitude
        - Lens metadata
        - Shutter timing

Environmental Requirements:

- Payload shall meet IP54 or higher
- Payload shall withstand vibration per American Society for Testing and Materials / Radio Technical Commission for Aeronautics (ASTM/RTCA) small UAS profiles
- Payload shall operate in light precipitation

Safety Requirements:

- Mechanical shutter shall be protected from dust and debris
- Payload shall include thermal protection for sensor overheating

Deliverables:

- Mechanical drawings (STEP, DXF)
- Electrical interface documentation
- Application Programming Interface / Software Development Kit (API/SDK) documentation
- Radiometric and geometric calibration certificates
- Lens calibration profiles
- Verification test results

TECHNICAL REQUIREMENT FOR MULTI-SENSOR IR/EO/LRF PAYLOAD:

This defines the desired technical, operational, environmental, and interface requirements for a next-generation, multi-sensor infrared/electro-optical/laser rangefinder (IR/EO/LRF) payload compatible with Freefly aircraft. The payload is intended to support public safety, search and rescue, wildfire, natural resource management, and general aerial inspection missions. The requirements herein represent target specifications for offeror’s development and evaluation.

Scope: This applies to the design, development, integration, and testing of a stabilized multi-sensor payload incorporating:

- High resolution zoom EO camera
- Wide angle EO camera
- Radiometric thermal infrared camera
- Laser range finder (LRF)
- Integrated near infra-red (NIR) illuminator
- 3 axis stabilized gimbal
- Freefly compatible mechanical and electrical interfaces
- Motion Imagery Standards Board (MISB)-compliant FMV metadata generation
- National Defense Authorization Act (NDAA) Compliance
- The payload must be fully compatible with Freefly payload interfaces

System Overview:

The payload shall provide day/night Intelligence, Surveillance, and Reconnaissance (ISR), mapping, thermal inspection, and ranging capabilities in a single integrated unit. The system shall support real time streaming, onboard recording, and full operator control of all sensors.

Functional Requirements:

- Complete installed payload(sensor + gimbal + mounting hardware + required accessories) weight shall be ≤ 1,100 g
- The payload shall consume ≤ 30 W with all sensors active
- The payload shall support IP54 or higher environmental protection
- The payload shall operate from −20° C to 50° C
- The payload shall store safely from −20° C to 50° C
- The payload shall mount using the Freefly Quick Detach Payload Interface

Gimbal Requirements:

- Stabilization
    - The gimbal shall provide 3 axis stabilization (tilt, roll, pan)
    - Stabilization accuracy shall be ≤ ±0.03° during flight
- Mechanical Range
    - Tilt: −130° to +40°
    - Roll: ±30°
    - Pan: ±330°
- Controllable Range
    - Tilt: −120° to +30°
    - Pan: ±320°

- Controllable Speed
    - Tilt: 100°/S
    - Pan: 100°/S
    - Roll: 100°/S

EO Zoom Camera Requirements:

- Sensor
    - Sensor shall be 1/1.8" Complementary Metal Oxide Semiconductor (CMOS) or larger
    - Effective resolution shall be ≥ 20 megapixel (MP)
- Optics
    - Optical zoom range shall be ≥ 20× with up to 240X digital
    - Focal length shall span 7–170 mm (or equivalent)
    - Aperture shall range f/1.6–f/16
    - Diagonal field of view (DFOV) shall range 65°–3°
- Imaging
    - Max photo resolution shall be ≥ 5400×3600 pixels
    - Video shall support 4K @ 30 fps and 1080p @ 30 fps
    - Low light modes shall support 1080p @ 25/15/5 fps
    - ISO range shall be 100–25600, with extended mode of at least 200,000

EO Wid-Angle Camera Requirements:

- Sensor
    - Sensor shall be 1/1.3" CMOS or larger
    - Effective resolution shall be ≥ 20 MP; ≥ 48 MP preferred
- Optics
    - Focal length shall be 24 mm (35-mm equivalent)
    - Maximum aperture shall be f/1.8 or faster (e.g., f/1.7)
    - DFOV shall be ≥ 82°
- Imaging
    - Max photo resolution shall be ≥ 4000×3000 pixels
    - Video shall support 4K @ 30 fps and 1080p @ 30 fps

Radiometric Thermal Infrared Camera Requirements:

- Detector
    - Detector shall be an uncooled VOx microbolometer
    - Resolution shall be ≥ 640×512; 1280×1024 preferred
    - Pixel pitch shall be 12 µm

- NETD shall be ≤ 50 mK; ≤ 40 mK preferred (at 30°C, F/1.0)
- Optics
  - Focal length shall be 24 mm (equivalent ~50 mm)
  - Aperture shall be f/0.95
  - Diagonal field of view (DFOV) shall be ~45°
  - Digital zoom shall be ≥ 8× while maintaining full frame rate
- Radiometry
  - The camera shall support full radiometric capture and overlay on the runtime video.
  - Temperature ranges:
    - High Gain: −20° C to 150° C
    - Low Gain: 0° C to 600° C
  - Optional density filters shall extend measurement to 1600° C
  - Radiometric accuracy shall be ±2°C or ±2% of reading (whichever is greater)
- Video
  - Thermal video shall support 30 fps
  - Video format shall be MP4 (H.264/H.265)
- Photo
  - Photo formats shall include JPEG and a radiometric format (TIFF or R-JPEG)

Laser Range Finder Requirements:

- Performance
  - Measurement range shall be 3–3000 m (for a 20% reflectivity target under standard atmospheric conditions)
  - Measurement accuracy shall be:
    - 3–500 m: ±(0.2 m + 0.15% of distance)
    - 500–3000 m: ±1.0 m
- Laser Characteristics
  - Wavelength shall be 905 nm
  - Laser shall comply with IEC 60825-1 Class 1 eye-safety limits for 905 nm

NIR Illuminator Requirements:

- Wavelength shall be 850 nm
- Full-angle FOV shall be ~4.6°, providing an illumination footprint of ~8 m diameter at 100 m
- Laser classification shall be Class 1

Software and Control Requirements:

- Integration
    - Payload shall be fully controllable via Freefly API/SDK
    - Payload shall support linked zoom between EO and I
    - Payload shall support click to aim and target handoff
    - Command latency shall be ≤ 150 ms for gimbal and zoom operations
- Recording
    - Payload shall record to UHS-I V60 SD card or higher
    - File system shall be exFAT
    - Optional AES 256 encryption shall be supported
    - Payload shall support simultaneous recording of EO, IR, and metadata streams.
- Full Motion Video (FMV) Metadata
    - The payload shall support real time and recorded Full Motion Video (FMV) with embedded Motion Imagery Standards Board (MISB) compliant metadata
    - Metadata shall conform to applicable MISB standards for airborne ISR systems (e.g., ST 0601, ST 0903, or successor standards)
    - MISB metadata shall include, at minimum:
        - Platform position (latitude/longitude/altitude)
        - Platform attitude (roll/pitch/yaw)
        - Sensor FOV (horizontal/vertical)
        - Sensor relative azimuth/elevation
        - Platform velocity (if available)Sensor position
        - Sensor attitude
        - Frame center coordinates
        - Time stamps
        - Field of view
        - Slant range (if available)
        - Ground sample distance (if applicable)
    - Metadata shall be synchronized with each video frame and embedded in the transport stream
    - The payload shall support MISB compliant metadata for both EO and IR video streams
    - The payload shall be compatible with common FMV exploitation tools used by public safety, SAR, and federal land management agencies
    - Metadata shall be compatible with oriented imagery workflows

Environmental Requirements:

- Payload shall meet IP54 or higher
- Payload shall withstand vibration per ASTM/RTCA small UAS profiles
- Payload shall withstand light precipitation during operation

Safety Requirements:

- All lasers shall be Class 1
- Thermal sensor shall include sunburn protection

Deliverables:

- Offerors shall provide:
  - Mechanical drawings (STEP, DXF)
  - Electrical interface documentation
  - API/SDK documentation
  - Environmental qualification results
  - Radiometric calibration certificates
  - Verification test results

TECHNICAL REQUIREMENT FOR THERMAL + MULTISPECTRAL IMAGING PAYLOAD:

This defines the desired technical, operational, environmental, and interface requirements for a Freefly compatible, multi sensor payload combining radiometric thermal, multispectral, and RGB imaging. The payload is intended for wildfire detection, SAR, structure fire overwatch, fuels and vegetation mapping, and land management analytics.

Scope: Applies to the design, development, integration, and testing of a stabilized payload incorporating:

- Radiometric thermal camera
- Multispectral camera (vegetation/fuels)
- RGB context camera
- Freefly compatible mechanical and electrical interfaces
- Real time live video feed to the pilot
- Onboard recording and metadata capture

The payload shall provide:

- High resolution radiometric thermal imaging
- Multispectral reflectance data for vegetation/fuels analysis
- Red, Green, Blue (RGB) context imagery
- Real time video streaming to the pilot
- Calibrated data products for downstream analysis

Functional Requirements:

- Complete installed payload(sensor + gimbal + mounting hardware + required accessories) weight shall be ≤ 1100 g.
- Power consumption shall be ≤ 30 W.
- Environmental protection shall be IP54 or higher.
- Operating temperature shall be −10° to 50° C.
- Payload shall mount using the Freefly Quick Detach Payload Interface.
- Payload shall be compatible with Freefly platforms.

Thermal Camera Requirements:

- Detector
  - Detector shall be an uncooled VOx microbolometer.
  - Resolution shall be 640×512 or higher.
  - Pixel pitch shall be 12 µm.
  - NETD shall be ≤ 50 mK.
- Radiometry
  - The camera shall support full radiometric capture.
  - Temperature accuracy shall be ±2°C or ±2%, whichever is greater.
  - Temperature measurement range shall be:
    - High Gain: −20° C to 150° C
    - Low Gain: 0° C to 600° C
- Optics
  - Focal length shall be 13–19 mm (or equivalent).
  - Aperture shall be f/1.0 or faster.
  - DFOV shall be ≥ 45°.
- Thermal Video
  - Thermal video shall support 30 fps.
  - Video format shall be MP4 (H.264/H.265).
  - Thermal video shall be available for real time pilot viewing.

Multispectral Camera Requirements:

- Bands
  - The multispectral camera shall support at least five discrete spectral bands, including:
    - Blue (~450 nm)
    - Green (~550 nm)
    - Red (~650 nm)
    - Red Edge (~720 nm)
    - Near Infrared (NIR, ~840 nm)
  - Band centers and bandwidths shall be documented and stable.
- Radiometric/Reflectance Performance
  - The multispectral system shall support reflectance calibrated output.
  - The system shall support calibration using a reflectance panel.

- The system shall include a downwelling light sensor or equivalent for irradiance normalization.
  - The system shall produce data suitable for standard vegetation indices (e.g., Normalized Difference Vegetation Index(NDVI), Normalized Difference Red Edge Index (NDRE), Green Normalized Difference Vegetation Index (GNDVI)).
- Spatial Resolution
  - Ground sample distance (GSD) at 120 m AGL shall be ≤ 10 cm for multispectral bands.
- Multispectral Capture
  - Multispectral capture shall support frame synchronized bands (all bands captured for the same ground footprint).
  - Capture interval shall be ≤ 1.0 s at full resolution.

RGB Context Camera Requirements:

- Sensor
  - Sensor shall be 20 MP or higher.
  - Sensor shall support low noise imaging for dusk/dawn operations.
- Optics
  - Lens shall provide a wide angle FOV suitable for mapping and context.
- Imaging
  - Max photo resolution shall be ≥ 5120×3840.
  - 7.3.2 Video shall support 1080p @ 30 fps.

Live Video Feed Requirements:

- Real-time Pilot Video
  - The payload shall provide continuous live video feed to the pilot for at least:
    - Thermal stream
    - RGB stream
  - Live video latency shall be ≤ 250 ms end to end under nominal link conditions.
  - Live video shall be viewable in Freefly Ground Control or equivalent GCS.
  - The system shall support pilot selectable view modes, including:
    - RGB only
    - Thermal only
    - Picture in picture or split view (if supported by hardware)
  - Live video shall remain stable during:
    - High-rate gimbal movement
    - High speed flight
    - Low light conditions

Data and Metadata Requirements:

- Storage
  - Payload shall record to UHS-I V60 SD card or higher.
  - File system shall be exFAT.
- File Types
  - Thermal stills shall be stored as radiometric JPEG (R JPEG) or equivalent.
  - Multispectral imagery shall be stored as band separated images or a multiband container with calibration metadata.
  - RGB stills shall be stored as minimally-compressed JPEG.
  - Video shall be stored as MP4.
- Metadata
  - Each image (thermal, multispectral, RGB) shall include:
    - GPS position
    - Altitude
    - Timestamp
    - Camera orientation
  - Thermal images shall include radiometric calibration data.
  - Multispectral images shall include:
    - Band identifiers
    - Calibration coefficients
    - Downwelling light sensor data (if used)

Software and Control Requirements:

- Integration
  - Payload shall be fully controllable via Freefly API/SDK.
  - Payload shall support:
    - Thermal gain mode selection
    - Thermal color palette selection
    - Multispectral capture control
    - RGB/thermal view switching
- Gimbal Control
  - Payload shall support gimbal pitch control from the pilot
  - Gimbal stabilization shall maintain $\leq \pm 0.03°$ error

Environmental Requirements:

- Payload shall meet IP54 or higher
- Payload shall withstand vibration per American Society for Testing and Materials (ASTM)/ Radio Technical Commission for Aeronautics (RTCA) small UAS profiles
- Payload shall operate in light precipitation

Safety Requirements:

- Thermal sensor shall include sunburn protection
- Payload shall fail safe on power loss

- Payload shall not exceed safe surface temperatures during operation

Deliverables:

- Offerors shall provide:
    - Mechanical drawings (STEP, DXF)
    - Electrical interface documentation
    - API/SDK documentation
    - Radiometric calibration certificates (thermal)
    - Reflectance calibration documentation (multispectral)
    - Environmental qualification results
    - Verification test results